Quantitative EEG and ERP report

CLINIC C LOGO

Patient:	Mock Patient 042
Date of birth:	1 Jan 2016
Age:	10 years
Sex:	female
Recorded:	15 Sep 2026
Case code:	MOCK-042
Earlier measurement:	12 Mar 2026

Software	V4 mockup 0.1
Norms	Mock norms 2026.1 (synthetic)
Recording system	Mitsar amplifier, WinEEG file (mockup label)
Report	R-MOCK-042 v1 · Draft, not issued

I. Summary

The largest marked deviations from the age norm in this measurement are: relative theta power, eyes open: O1 (z = -4.3), Pz (z = -4.2), O2 (z = -4.2); relative theta power, eyes closed: O2 (z = -4.3), O1 (z = -4.2), Pz (z = -4.1); SMR asymmetry index, eyes open: C3-C4 (z = 4.2); SMR asymmetry index, eyes closed: C3-C4 (z = 4.2); beta1 coherence, eyes closed: P3-O1 (z = 4.2), T3-T5 (z = 4.1), T5-O1 (z = 4.1), T3-C3 (z = 3.9), C3-T5 (z = 3.9), C3-P3 (z = 3.9), T5-P3 (z = 3.8); theta coherence, eyes open: C3-T4 (z = -3.7); beta1 coherence, eyes open: C3-T5 (z = 3.7), C3-P3 (z = 3.7), T5-O1 (z = 3.7), P3-O1 (z = 3.7), T3-C3 (z = 3.6), T3-T5 (z = 3.6), T5-P3 (z = 3.6); P1 latency, Go: O1 (z = 3.7), O2 (z = 3.2). SUM-2

This list ranks the marked values of these analyses by |z|: absolute and relative band power, maps per 1 Hz band, alpha peak frequency, theta/beta ratio, frontal alpha asymmetry index, asymmetry map, coherence per electrode pair, ERP peak amplitude and latency, and VCPT task results. It shows the 5 largest and every marked value of 3 z or more. Not ranked: the spectra per 0.5 Hz bin, which are shown as curves against the norm, and global coherence, which is still to be confirmed. SUM-RULE-2

Marked values in this measurement: band power 6 of 456, maps per 1 Hz band 0 of 1102, alpha peak and ratios 0 of 6, asymmetry map 2 of 96, coherence 15 of 2052, ERP peaks 2 of 21 and VCPT task 0 of 4. SUM-COUNT-1

II. Signal quality and processing

Eyes open: 164 of 180 seconds were analysed after artefact removal. Q-DUR-1

Eyes closed: 171 of 180 seconds were analysed after artefact removal. Q-DUR-1

VCPT: 1168 of 1220 seconds were analysed after artefact removal. Q-DUR-1

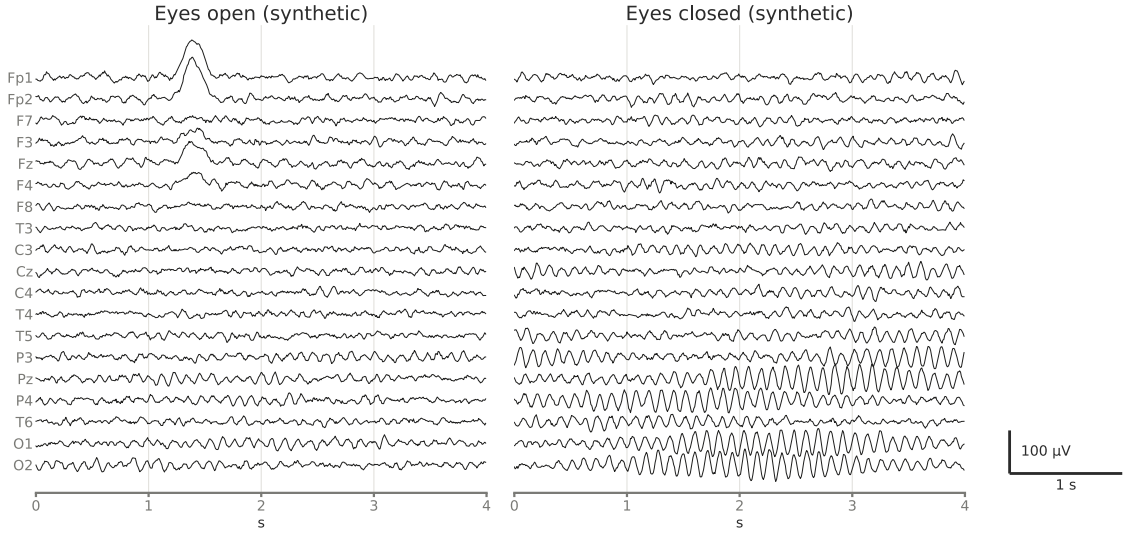
Accepted VCPT trials: Go 95, NoGo 92, Ignore 93, Novelty 95 (of 100 each). Q-TRIALS-1

Artefact removal changed by the clinician: component 6 (muscle, left temporal) was kept although the automatic cleaning proposed removing it; 2 of 19 components were removed. Q-ICA-MAN-1

Analysed time: eyes open 164 s, eyes closed 171 s, VCPT 1168 s; segments confirmed by the clinician. Q-SEG-1

Alpha peak frequency marker, eyes closed, confirmed by the clinician at 9.2 Hz. MK-1

III. Raw EEG excerpt



Synthetic signal, not a recording: the first 4 s of the analysed segments, eyes open and eyes closed, average reference, 19 channels. Scale bars: 100 µV and 1 s.

IV. Anamnesis

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

V. Spectra and maps

Eyes open

Figure 1: Absolute power, z against the age norm, Eyes open

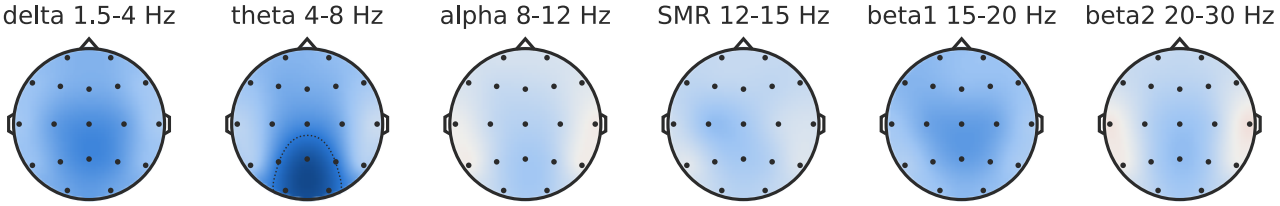
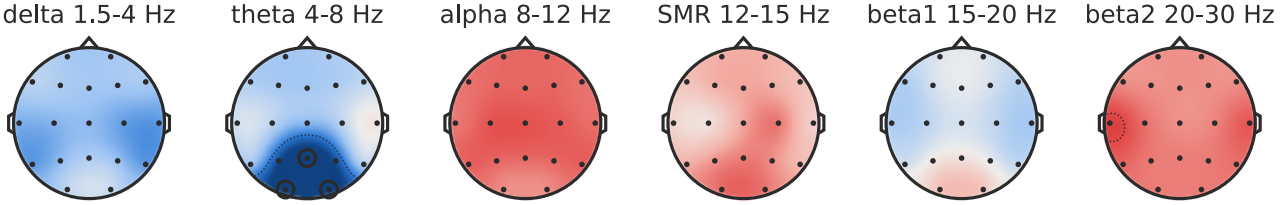


Figure 2: Relative power, z against the age norm, Eyes open



Relative theta power, eyes open, is below the age norm at Pz (z = -4.2), O1 (z = -4.3) and O2 (z = -4.2). SP-REL-1

The alpha peak frequency, eyes open, is 9.4 Hz and is not marked (z = 0.4). APF-IN-2

Eyes closed

Figure 3: Absolute power, z against the age norm, Eyes closed

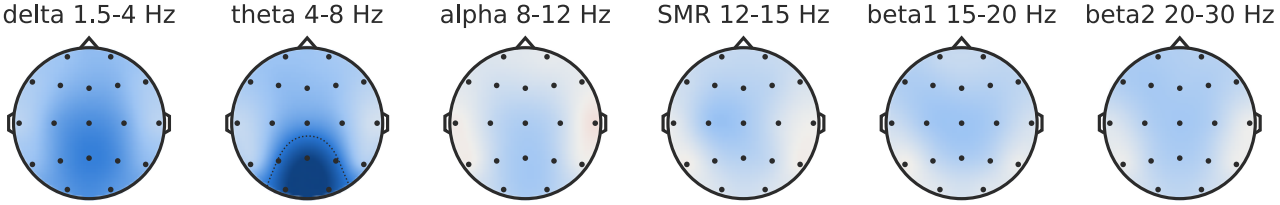
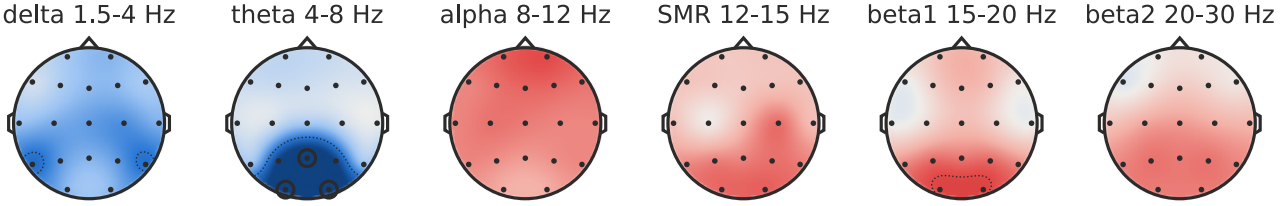
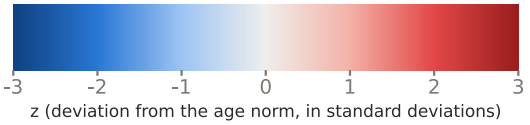


Figure 4: Relative power, z against the age norm, Eyes closed



Relative theta power, eyes closed, is below the age norm at Pz (z = -4.1), O1 (z = -4.2) and O2 (z = -4.3). SP-REL-1

The alpha peak frequency, eyes closed, is 9.2 Hz and is not marked (z = 0.0). APF-IN-2



Ring: marked value (after correction for multiple comparisons).

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

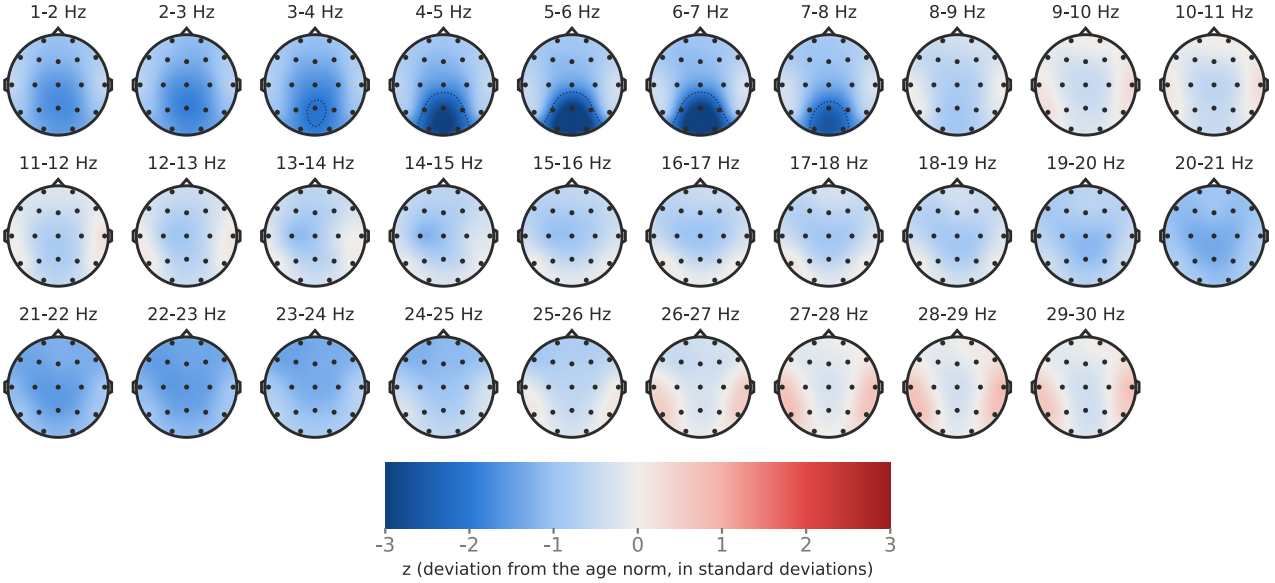
CLINIC C LOGO

Clinic C · [Street placeholder] · [Postcode and town placeholder] · [Country placeholder]
[web address placeholder]

VI. Maps per 1 Hz band (absolute power, z)

Eyes closed

Figure 5: Maps per 1 Hz band (absolute power, z), Eyes closed

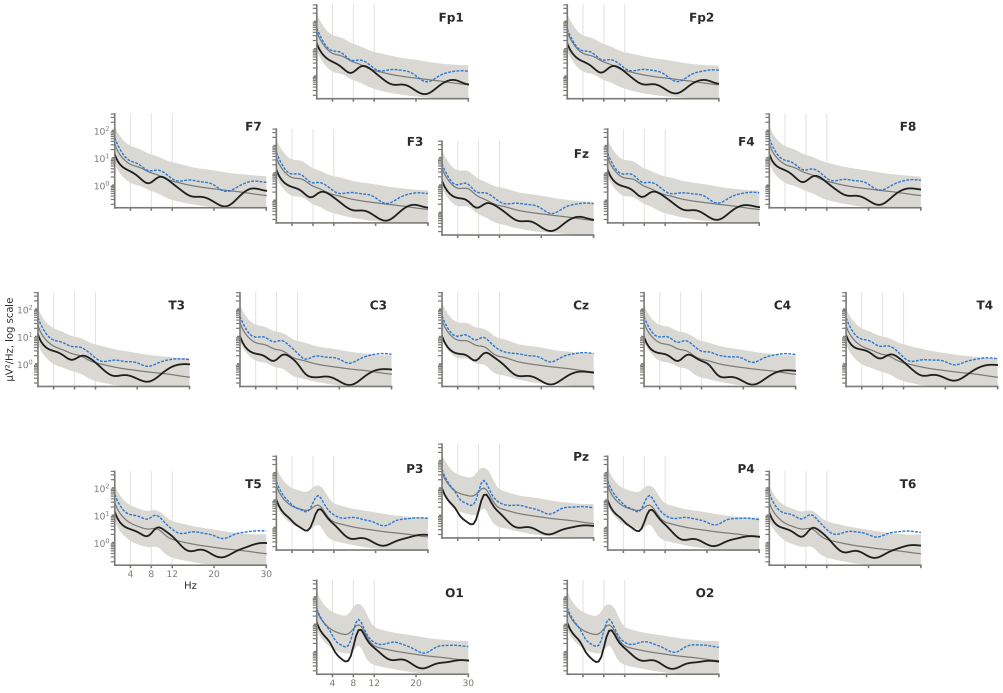


Ring: marked value (after correction for multiple comparisons).

VII. Absolute spectra per electrode

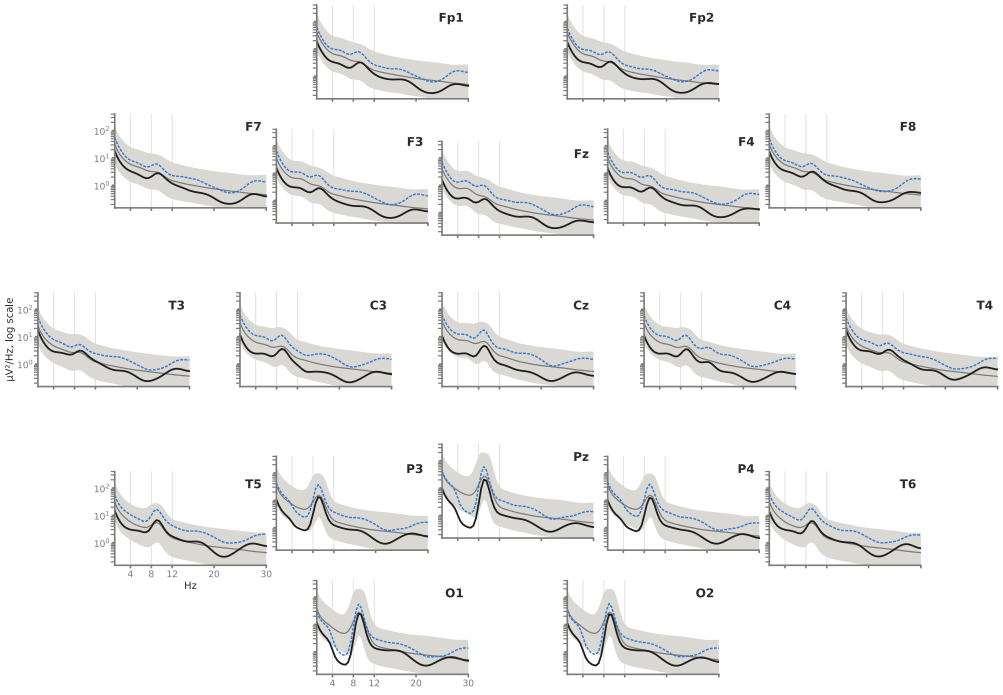
Eyes open

Figure 6: Absolute spectra per electrode, Eyes open



Eyes closed

Figure 7: Absolute spectra per electrode, Eyes closed



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Blue dashed: earlier measurement.

CLINIC C LOGO

Clinic C · [Street placeholder] · [Postcode and town placeholder] · [Country placeholder]
[web address placeholder]

VIII. Band ratios and indices

Measure		Value	Norm mean	z
Theta/beta ratio at Cz	Eyes open	2.35	2.76	-0.5
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes open	0.01	-0.01	0.2
Alpha peak frequency (Hz)	Eyes open	9.4	9.1	0.4
Theta/beta ratio at Cz	Eyes closed	2.00	2.82	-1.0
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes closed	0.05	-0.01	0.6
Alpha peak frequency (Hz)	Eyes closed	9.2	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

The theta/beta ratio at Cz, eyes open, is 2.35 (z = -0.5). TBR-1

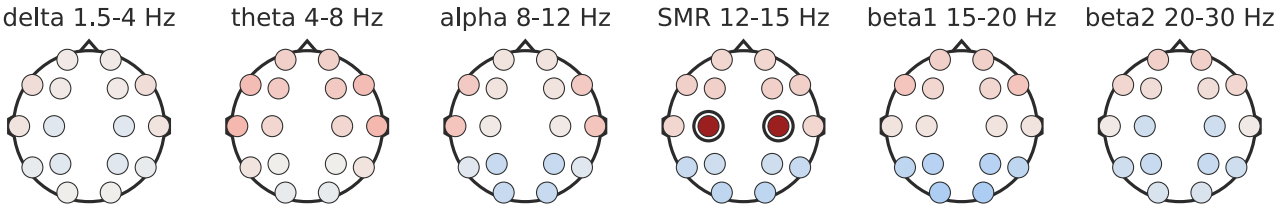
The frontal alpha asymmetry index (ln F4 minus ln F3), eyes open, is 0.01 (z = 0.2). FAA-1

The theta/beta ratio at Cz, eyes closed, is 2.00 (z = -1.0). TBR-1

The frontal alpha asymmetry index (ln F4 minus ln F3), eyes closed, is 0.05 (z = 0.6). FAA-1

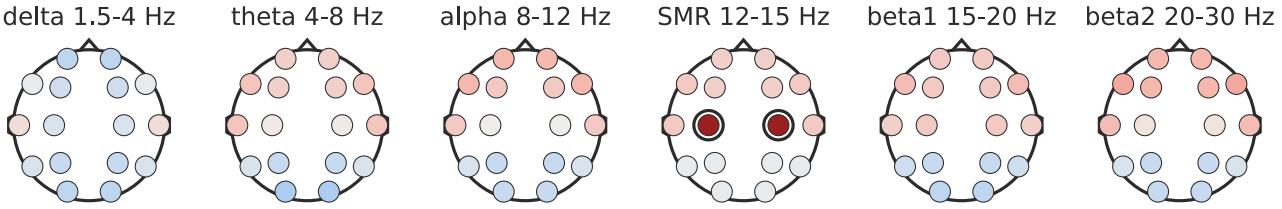
IX. Asymmetry

Eyes open

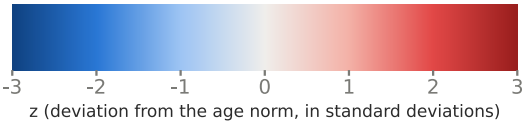


The SMR asymmetry index, eyes open, is above the age norm for C3-C4 (z = 4.2). ASY-1

Eyes closed



The SMR asymmetry index, eyes closed, is above the age norm for C3-C4 (z = 4.2). ASY-1



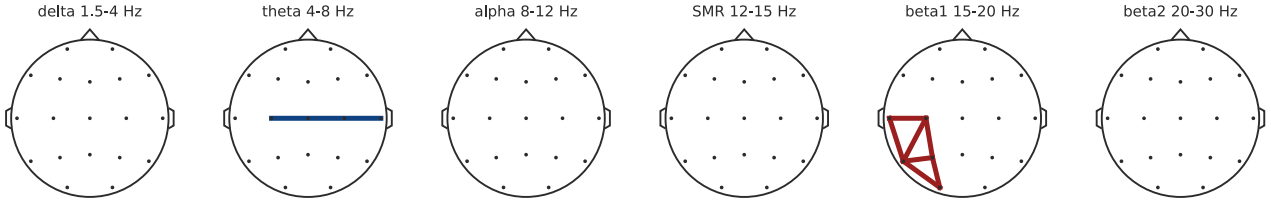
Asymmetry index (right minus left) / (right plus left) per electrode pair, z against the age norm. Ring: marked value (after correction for multiple comparisons).

CLINIC C LOGO

Clinic C · [Street placeholder] · [Postcode and town placeholder] · [Country placeholder]
[web address placeholder]

X. Coherence

Eyes open



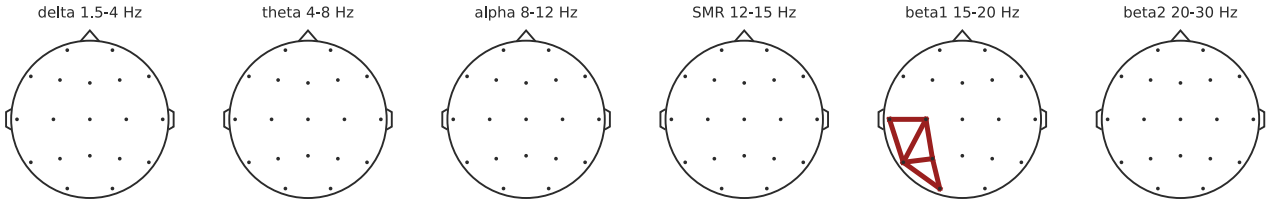
Theta coherence, eyes open, is below the age norm for 1 electrode pair: C3-T4 (z = -3.7). COH-1

Beta1 coherence, eyes open, is above the age norm for 7 electrode pairs: C3-T5 (z = 3.7), C3-P3 (z = 3.7), T5-O1 (z = 3.7), P3-O1 (z = 3.7), T3-C3 (z = 3.6), T3-T5 (z = 3.6) and 1 more in the table. COH-1

Figure 8: Marked electrode pairs, Eyes open

Band	theta	beta1					
Pair	C3-T4	C3-T5	C3-P3	T5-O1	P3-O1	T3-C3	T3-T5	T5-P3
Coherence	0.01	0.59	0.64	0.61	0.69	0.65	0.60	0.70
z	**-3.7**	**3.7**	**3.7**	**3.7**	**3.7**	**3.6**	**3.6**	**3.6**

Eyes closed



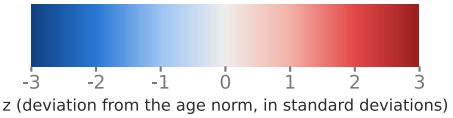
Beta1 coherence, eyes closed, is above the age norm for 7 electrode pairs: P3-O1 (z = 4.2), T3-T5 (z = 4.1), T5-O1 (z = 4.1), T3-C3 (z = 3.9), C3-T5 (z = 3.9), C3-P3 (z = 3.9) and 1 more in the table. COH-1

Figure 9: Marked electrode pairs, Eyes closed

Band	beta1						
Pair	P3-O1	T3-T5	T5-O1	T3-C3	C3-T5	C3-P3	T5-P3
Coherence	0.73	0.64	0.64	0.67	0.60	0.65	0.71
z	**4.2**	**4.1**	**4.1**	**3.9**	**3.9**	**3.9**	**3.8**

Global coherence (mean over all pairs), to confirm

	delta	theta	alpha	SMR	beta1	beta2
Eyes open, Value	0.20	0.20	0.23	0.22	0.22	0.19
Eyes open, z	-1.1	-1.1	-0.4	-0.1	0.3	-0.1
Eyes closed, Value	0.19	0.21	0.26	0.22	0.21	0.17
Eyes closed, z	-1.5	-0.9	-0.1	0.0	0.1	-0.7



Lines: marked electrode pairs, coloured by z. Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z. Global coherence is a core function still to be confirmed; it is not ranked in the summary.

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

XI. Event-related potentials at the electrodes (channel ERPs)

Figure 10: Event-related potentials at the electrodes (channel ERPs): Go

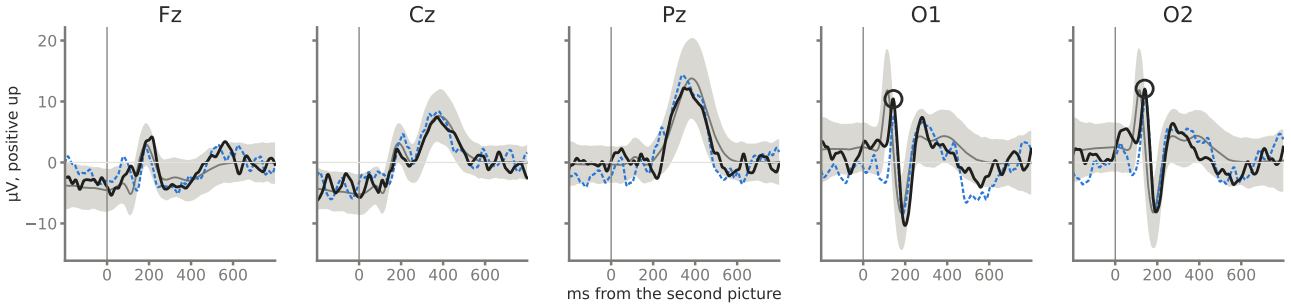
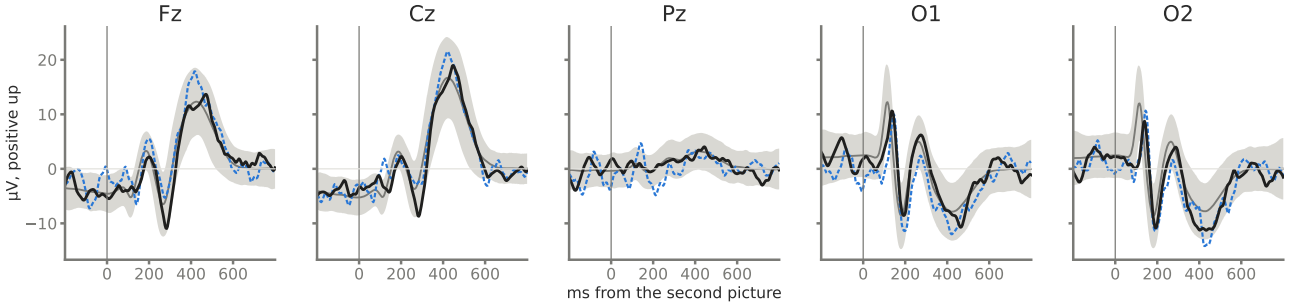


Figure 11: Event-related potentials at the electrodes (channel ERPs): NoGo



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Time 0: second stimulus. Ring: marked value (after correction for multiple comparisons). Blue dashed: earlier measurement.

Figure 12: ERP peak values

Component	Trial type	Electrode	Amplitude (µV)	z	Latency (ms)	z
P1	Go	O1	10.4	-0.8	144	**3.7**
P1	Go	O2	12.1	-0.3	140	**3.2**
N1	Go	O1	-10.3	-0.3	200	1.6
N1	Go	O2	-8.1	0.4	192	1.0
N2	NoGo	Fz	-11.0	-1.2	284	1.2
N2	NoGo	Cz	-8.7	-1.7	284	1.4
P3	Go	Pz	12.2	-0.9	348	-1.3
P3	NoGo	Cz	19.0	0.2	448	0.8
P3	NoGo	Fz	13.7	0.0	472	1.5
P3a	Novelty	Cz	17.4	0.7	364	1.4
CNV	Go	Cz	-3.9	0.7	-200 to 0	

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

ERP peaks confirmed by the clinician. ERP-PEAK-1

P1 latency at O1, Go trials, is later than the age norm (144 ms, z = 3.7). ERP-LAT-1

P1 latency at O2, Go trials, is later than the age norm (140 ms, z = 3.2). ERP-LAT-1

Clinician's comment

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

CLINIC C LOGO

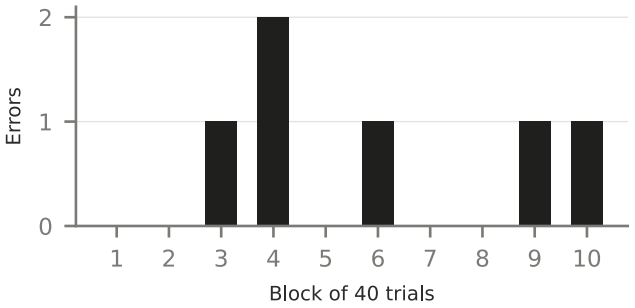
Clinic C · [Street placeholder] · [Postcode and town placeholder] · [Country placeholder]
[web address placeholder]

XII. VCPT task performance

Measure	Value	Percentile	z
Omission errors	2	46	-0.1
Commission errors	4	81	0.9
Mean reaction time	409 ms	47	-0.1
Reaction time variability (SD)	86 ms	42	-0.2

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 13: Errors per block of 40 trials (no norm)



Omission errors: 2, not marked (percentile 46). BEH-IN-2

Commission errors: 4, not marked (percentile 81). BEH-IN-2

Mean reaction time: 409 ms, not marked (percentile 47). BEH-IN-2

Reaction time variability (SD): 86 ms, not marked (percentile 42). BEH-IN-2

XIII. Comparison of measurements

Measure	Earlier 12 Mar 2026	z	Current 15 Sep 2026	z
Omission errors	3	0.3	2	-0.1
Commission errors	5	1.2	4	0.9
Mean reaction time	421 ms	0.1	409 ms	-0.1
Reaction time variability (SD)	95 ms	0.2	86 ms	-0.2
Theta/beta ratio at Cz, Eyes open	2.31	-0.6	2.35	-0.5
Alpha peak frequency (Hz), Eyes open	9.0	-0.1	9.4	0.4
Theta/beta ratio at Cz, Eyes closed	2.32	-0.6	2.00	-1.0
Alpha peak frequency (Hz), Eyes closed	9.1	-0.1	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 14: Earlier 12 Mar 2026: Absolute power, z against the age norm, Eyes closed

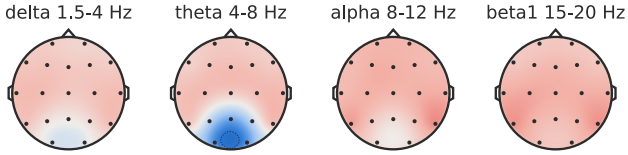
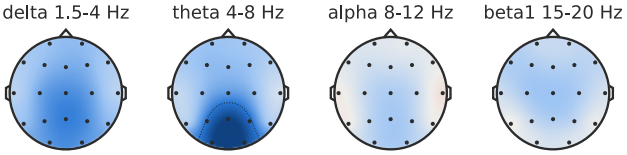


Figure 15: Current 15 Sep 2026: Absolute power, z against the age norm, Eyes closed



Ring: marked value (after correction for multiple comparisons).

Beta1 coherence T5-P3, eyes closed: z = 4.5 on 12 Mar 2026, z = 3.8 on 15 Sep 2026. CMP-1

Beta1 coherence T3-C3, eyes closed: z = 4.4 on 12 Mar 2026, z = 3.9 on 15 Sep 2026. CMP-1

Relative theta power at O1, eyes open: z = -3.8 on 12 Mar 2026, z = -4.3 on 15 Sep 2026. CMP-1

Relative theta power at O2, eyes closed: z = -3.7 on 12 Mar 2026, z = -4.3 on 15 Sep 2026. CMP-1

Beta1 coherence P3-O1, eyes closed: z = 4.3 on 12 Mar 2026, z = 4.2 on 15 Sep 2026. CMP-1

Relative theta power at Pz, eyes open: z = -3.6 on 12 Mar 2026, z = -4.2 on 15 Sep 2026. CMP-1

The earlier measurement of 12 Mar 2026 was processed again with the current software and norms (V4 mockup 0.1; Mock norms 2026.1 (synthetic)). L-REPROC-1

Measurements are shown side by side, each against its own age norm; differences between them are not tested for statistical significance. L-SIDE-1

XIV. Notes on the results

The results are one input to the clinician's assessment. The software does not diagnose and does not recommend treatment. L-AID-1

z is the distance of a value from the normative mean, in standard deviations of the normative sample; about 95% of the normative sample lies between z = -2 and z = 2. L-Z-1

Marked values differ from the age norm after correction for multiple comparisons, which keeps chance results among the many comparisons rare; a value beyond 2 z that is not marked may deviate by chance alone. The method of correction is still to be chosen. L-MC-1

XV. Interpretation

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

XVI. Conclusion

Placeholder for the clinician's text. Written in the V4 report editor; the system does not write here.

Issued by

Issued by	_____ (entered when the clinician issues the report)
Date of issue	_____ (entered when the clinician issues the report)
Version	1 · Draft, not issued
Replaces	none, first version

CLINIC C LOGO

Clinic C · [Street placeholder] · [Postcode and town placeholder] · [Country placeholder]
[web address placeholder]